

EventZen Event Management System

System Architecture and ERD Documentation

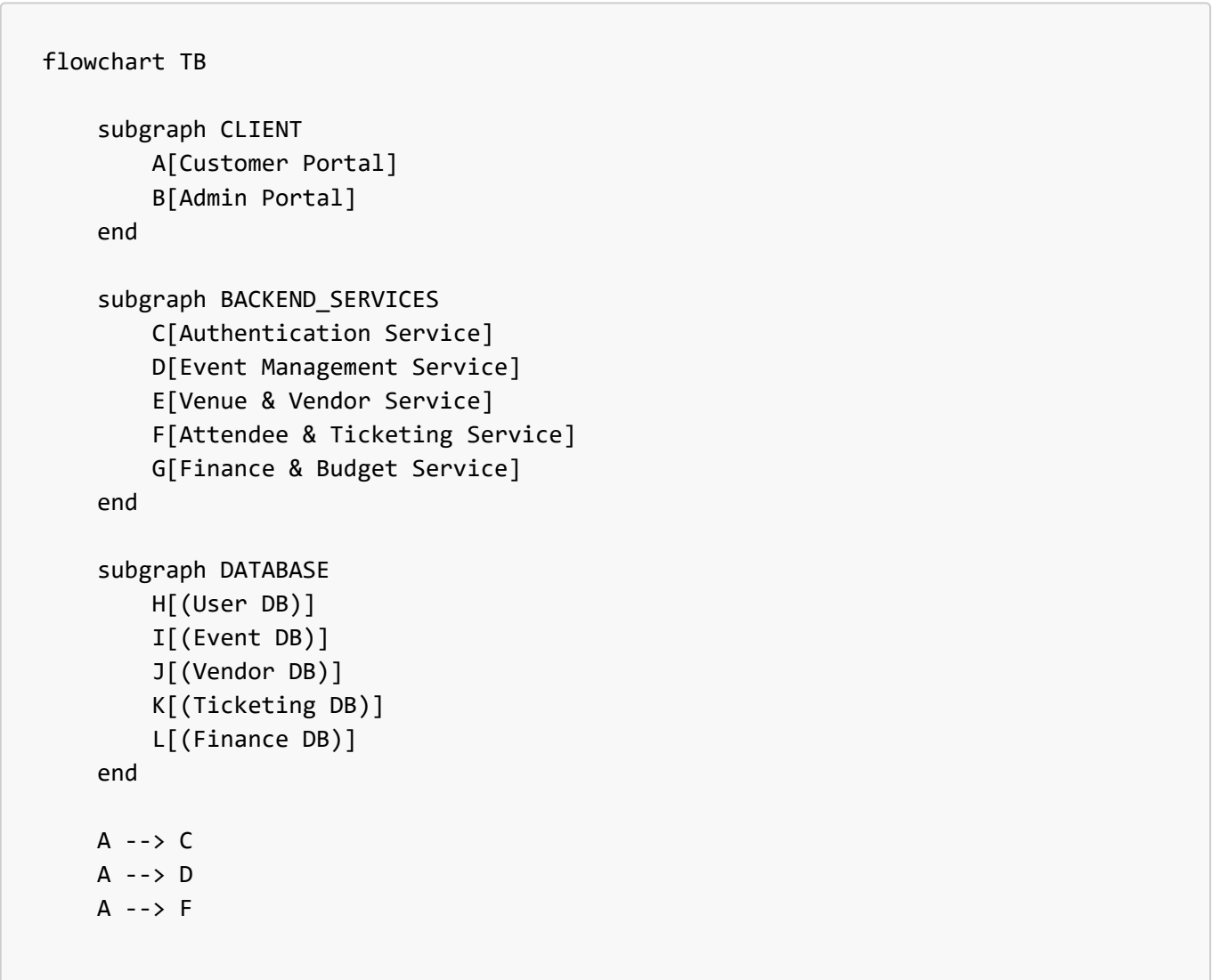
This document presents the **system architecture and database design** for the **EventZen Event Management System**. The architecture is designed to support scalable event planning operations including event scheduling, vendor coordination, attendee registration, and financial management.

The system is organized into modular components to improve maintainability, scalability, and clarity of the system design.

The architecture includes:

- Top-level system architecture
- Five functional modules
- Clearly defined service boundaries
- Logical grouping of database entities

System Architecture



```
B --> C
B --> D
B --> E
B --> G

C --> H
D --> I
E --> J
F --> K
G --> L
```

This architecture separates the application into **frontend portals, backend services, and service-specific databases**, enabling independent scaling and modular development.

Modules

The system is divided into five primary modules:

1. Authentication & User Management
2. Event Management
3. Venue & Vendor Management
4. Attendee & Ticketing
5. Finance & Budget Management

Each module is expandable below to view its detailed **ER diagram**.

► Authentication & User Management Module

This module manages **users, authentication, and role-based authorization**.

```
erDiagram

    USER {
        UUID user_id PK
        string first_name
        string last_name
        string email
        string password_hash
        string phone
        boolean is_active
        datetime created_at
    }

    ROLE {
        UUID role_id PK
        string role_name
        string description
    }
```

```
PERMISSION {
    UUID permission_id PK
    string permission_name
    string module
    string action
}

USER_ROLE {
    UUID id PK
    UUID user_id FK
    UUID role_id FK
}

ROLE_PERMISSION {
    UUID id PK
    UUID role_id FK
    UUID permission_id FK
}

USER ||--o{ USER_ROLE : assigned
ROLE ||--o{ USER_ROLE : contains
ROLE ||--o{ ROLE_PERMISSION : grants
PERMISSION ||--o{ ROLE_PERMISSION : defines
```

► Event Management Module

This module manages **event creation, scheduling, and agenda management**.

erDiagram

```
EVENT {
    UUID event_id PK
    UUID organizer_id FK
    UUID category_id FK
    string event_title
    string event_type
    string description
    datetime start_time
    datetime end_time
    integer expected_attendees
    decimal estimated_budget
    string status
}

EVENT_CATEGORY {
    UUID category_id PK
    string category_name
}

EVENT_SESSION {
```

```

        UUID session_id PK
        UUID event_id FK
        string session_title
        string speaker_name
        datetime start_time
        datetime end_time
    }

    EVENT_AGENDA {
        UUID agenda_id PK
        UUID event_id FK
        string agenda_title
        datetime start_time
        datetime end_time
    }

    EVENT_CATEGORY ||--o{ EVENT : categorizes
    EVENT ||--o{ EVENT_SESSION : contains
    EVENT ||--o{ EVENT_AGENDA : includes

```

► Venue & Vendor Management Module

This module handles **venue reservations and vendor service management**.

erDiagram

```

VENUE {
    UUID venue_id PK
    string venue_name
    string address
    string city
    integer capacity
    decimal price_per_day
}

VENUE_HALL {
    UUID hall_id PK
    UUID venue_id FK
    string hall_name
    integer capacity
}

VENDOR {
    UUID vendor_id PK
    string vendor_name
    string service_type
    string email
    string phone
    decimal rating
}

```

```

EVENT_VENDOR {
    UUID contract_id PK
    UUID event_id FK
    UUID vendor_id FK
    decimal agreed_price
    string contract_status
}

EVENT_VENUE_BOOKING {
    UUID booking_id PK
    UUID event_id FK
    UUID venue_id FK
    datetime booking_start
    datetime booking_end
}

VENUE ||--o{ VENUE_HALL : contains
VENUE ||--o{ EVENT_VENUE_BOOKING : reserved
VENDOR ||--o{ EVENT_VENDOR : hired_for

```

► Attendee & Ticketing Module

This module manages **attendee registrations, ticket purchases, and check-in tracking**.

erDiagram

```

ATTENDEE {
    UUID attendee_id PK
    string first_name
    string last_name
    string email
    string phone
}

TICKET_TYPE {
    UUID ticket_type_id PK
    UUID event_id FK
    string ticket_name
    decimal price
    integer total_quantity
}

EVENT_REGISTRATION {
    UUID registration_id PK
    UUID event_id FK
    UUID attendee_id FK
    UUID ticket_type_id FK
    datetime registration_date
    string status
}

```

```

CHECKIN_LOG {
    UUID checkin_id PK
    UUID registration_id FK
    datetime checkin_time
    string gate
}

ATTENDEE ||--o{ EVENT_REGISTRATION : registers
TICKET_TYPE ||--o{ EVENT_REGISTRATION : selected
EVENT_REGISTRATION ||--o{ CHECKIN_LOG : logs

```

► Finance & Budget Management Module

This module manages **budget planning, payments, and expense tracking**.

erDiagram

```

BUDGET {
    UUID budget_id PK
    UUID event_id FK
    decimal estimated_total
    decimal approved_total
    decimal actual_total
}

BUDGET_ITEM {
    UUID item_id PK
    UUID budget_id FK
    string category
    decimal estimated_amount
    decimal actual_amount
}

PAYMENT {
    UUID payment_id PK
    UUID registration_id FK
    decimal amount
    string payment_method
    string payment_status
    datetime payment_date
}

EXPENSE {
    UUID expense_id PK
    UUID event_id FK
    decimal amount
    string category
    date expense_date
}

BUDGET ||--o{ BUDGET_ITEM : contains

```

```
EVENT ||--|| BUDGET : allocated
EVENT ||--o{ EXPENSE : incurs
EVENT_REGISTRATION ||--o{ PAYMENT : generates
```

Final System Structure

Module	Purpose
Authentication	User authentication, roles, and permissions
Event Management	Event creation, sessions, and scheduling
Venue & Vendor	Venue booking and vendor services
Attendee & Ticketing	Registration, ticketing, and check-ins
Finance & Budget	Budget tracking, payments, and expenses

Design Characteristics

The proposed system design demonstrates:

- Modular architecture
- Scalable microservice-oriented structure
- Normalized database schema
- Clear separation of concerns
- Real-world event management workflow support